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# Isolation Is Not Confidentiality: Security Boundaries of Geometric Access Control in RAG

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**Abstract.** Retrieval-Augmented Generation (RAG) systems increasingly rely on vector databases containing documents governed by different access-control policies. Recent search-preserving geometric approaches enforce such access control by structuring the embedding space so that transformed embeddings, which we call protected representations, remain retrievable without exposing information across policy boundaries.
We provide a unified security analysis of this paradigm, covering policy isolation, representation confidentiality, query forgery resistance, and policy-membership hiding. We consider both a curious cloud service provider observing the protected representations stored in the vector database and a curious user holding aligned pairs of original embeddings and protected representations. Evaluating the available implementations of the works proposing these geometric transformation mechanisms across three encoders, we find that geometric transformations preserve retrieval utility and provide strong policy isolation, but do not provide the other security properties. A curious user can recover the original embeddings from a few hundred pairs and forge policy-compatible queries at a higher cost that grows with encoder dimension for secret per-policy transformations. Moreover, policy-dependent geometry can leak policy membership even when unauthorized retrieval is prevented. These results show that policy isolation, confidentiality, forgery resistance, and membership hiding are distinct security objectives.

**Keywords:** Retrieval-Augmented Generation · Access Control · Embedding Security · Geometric Transformations · Vector Retrieval · Privacy

## 1 Introduction

Retrieval-Augmented Generation (RAG) is more and more considered to ground large language models (LLMs) with external knowledge, which is encoded as embeddings by an encoder (a model that maps text to numeric vectors) and stored in a latent embedding space to enable similarity search between queries and knowledge [1]. In enterprise deployments, a vector database may contain documents governed by different access-control policies [2]. This creates a mismatch: embedding similarity search determines which documents are relevant to a query, but not whether the user is authorized to access them. As a result,

semantically similar documents governed by different policies may be retrieved together unless authorization is enforced separately.

Existing approaches address this problem using metadata-based filtering [21], policy-specific indexes [23], or cryptographic protection [26, 29]. However, these approaches either enforce authorization outside the embedding space, require separate storage structures, or apply additional mechanisms to perform similarity search over encrypted embeddings. A different approach is to encode authorization directly into embeddings by means of geometric transformations that preserve similarity. We refer to this paradigm as *search-preserving geometric access control*. Each policy is associated with a secret transformation of the embedding space; the transformed embedding, which we call the *protected representation*, is what the vector database stores, and authorized queries undergo the same transformation. Recent systems implement this approach using policy-specific transformations of stored embeddings [8–10]. With policy separation built into the embedding space, these approaches can enforce access control directly through similarity search while preserving retrieval utility.

However, preserving retrieval does not guarantee other security properties. Policy isolation and representation confidentiality are distinct objectives. A transformation may prevent cross-policy retrieval during normal operation while still being learnable from informative observations. Moreover, the same structure used to separate policies may reveal policy membership. This raises a broader question: what security properties does search-preserving geometric access control actually provide?

We study this question through similarity preservation and four security properties: policy isolation, representation confidentiality, query forgery resistance, and policy-membership hiding. We consider two adversaries. A curious cloud service provider has access to protected representations but no policy keys or original embeddings for the target policy. A curious user additionally holds aligned pairs of original embeddings and protected representations for a target policy. We use this case to evaluate the residual protection once enough observations are available to learn a policy-specific transformation. Our goal is therefore not to claim a cryptographic break of a particular system, but to characterize the security boundary of the search-preserving geometric access-control paradigm.

Our evaluation leads to three main findings. First, policy-specific geometric transformations can preserve retrieval utility while providing strong policy isolation. Second, isolation does not imply confidentiality or forgery resistance: aligned observations make the evaluated transformations learnable, so a curious user can recover the original embeddings from a few hundred pairs and forge policy-compatible queries at a higher, dimension-dependent cost. Third, geometric separation can leak policy membership, even when policy isolation is preserved.

*Contributions.* Our contributions are threefold:

- We provide a unified security characterization of geometric access control for multi-policy RAG, treating retrieval utility and four security properties

(policy isolation, representation confidentiality, query forgery resistance, and policy-membership hiding) as distinct objectives.
- We analyze the learnability of policy transformations from aligned pairs of original embeddings and protected representations, and distinguish representation confidentiality (recovering the original embedding) from query forgery resistance (constructing a policy-compatible query), two objectives that target different objects and need not hold together.
- We evaluate representative published mechanisms (policy-specific rotations, key-parametrized dimension augmentation, and distance-comparison-preserving encryption) and their fusion, on the authors' own implementations across three encoders. A curious user recovers the original embeddings from a few hundred aligned pairs and forges policy-compatible queries at a higher, dimension-dependent cost, and policy-dependent geometric separation exposes policy membership, including for the cryptographic scheme.

*Outline.* Section 2 reviews existing approaches to access control in RAG and situates search-preserving geometric access control among them. Section 3 defines the system and adversary models. Section 4 presents the security analysis of the four properties under a generic transformation model. Section 5 reports the experimental evaluation of the three mechanisms and their fusion across three encoders. Section 6 discusses implications and limitations, and Section 7 concludes.

## 2 Background and Related Work

### 2.1 Access Control in RAG

Access control in RAG systems is commonly enforced outside the embedding space. Metadata-based filtering associates documents or chunks with authorization attributes and restricts retrieval to those accessible to the user [21, 22, 27]. Policy-specific indexes provide a similar separation by maintaining separate retrieval structures for different authorization domains [23], and LLM-driven or multitenant gateways enforce authorization outside the vector database [24, 25]. These approaches provide explicit access-control semantics but require authorization information to remain available during retrieval.

Recent work has explored integrating access policies directly into the embedding space [8–10]. Trans-RAG applies policy-dependent transformations to embeddings so that queries and documents under the same policy remain compatible while different policies are separated [8]; other schemes use policy-specific rotations [9] or key-parametrized dimension augmentation [10]. This design reduces reliance on metadata filtering during vector search and allows multiple policies to coexist within the same retrieval infrastructure. Our work focuses on the security properties of search-preserving geometric access control rather than on access-control enforcement alone.

### 2.2 Search-Preserving Protection of Embeddings

Related work studies transformations that protect vector representations while preserving the similarity relationships needed for retrieval. Distance-comparison-preserving encryption (DCPE) [11] and related constructions [13] aim to hide the underlying representation while allowing a server to perform similarity-based operations. Other approaches obscure embeddings while preserving retrieval utility, using embedding-space alignment [14] or cell-keyed residual splitting [15].

This creates an important tension. A transformation must preserve enough structure for similarity search, but hiding that structure is precisely what makes the original embedding difficult to recover. In particular, deterministic transformations can become learnable from aligned pairs of original embeddings and protected representations. Classical results already show that distance- and property-preserving transformations are vulnerable to known-plaintext and inference attacks that recover the transformation or the underlying data [16, 17, 20, 30]. Unsupervised methods can even align embedding spaces without paired supervision [18, 19], so aligned observations are a sufficient but not a necessary condition for learning such a mapping. We extend this idea to the policy-specific RAG setting and, beyond reconstruction, to query forgery and policy-membership exposure. We therefore treat similarity preservation and representation confidentiality as separate properties rather than assuming that one implies the other.

### 2.3 Representation Leakage and Inversion Attacks

Learned embeddings and intermediate representations can leak information about their inputs. Prior work has demonstrated reconstruction and inversion attacks that recover inputs from such vectors, including attacks based on corresponding input–representation pairs [32–34]. Related attacks recover data from the access and query-leakage patterns of encrypted nearest-neighbor search [31]. These results motivate evaluating what an adversary can learn beyond the intended retrieval operation.

Our setting differs from conventional embedding inversion because the protected representation is generated by a policy-specific transformation. This leads to two related but distinct attack objectives. An adversary may either (i) recover the original embedding from a protected representation, or (ii) learn the transformation well enough to construct a policy-compatible query. Query forgery does not require recovering the original embedding and therefore cannot be inferred from reconstruction performance alone.

### 2.4 Positioning of This Work

Prior work has studied policy-aware retrieval transformations and embedding inversion attacks separately. RAG privacy surveys [35] and provably secure designs [28] discuss related risks but do not examine search-preserving geometric access control. We bridge these gaps by jointly analyzing policy isolation, representation confidentiality, query forgery resistance, and policy-membership hiding. Table 1 summarizes this positioning.

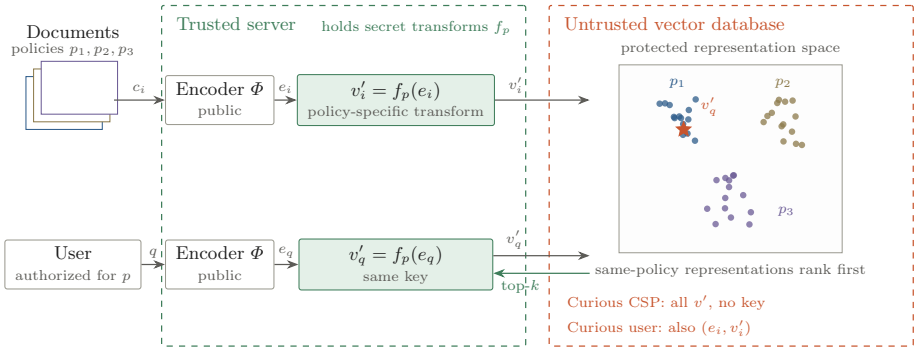


**Fig. 1.** System and threat model. Document and query embeddings from the public encoder $\Phi$ are transformed by a policy-specific secret transformation $f_p$ inside the trusted server; only the protected representations $v'$ reach the untrusted vector database, where same-policy representations rank first. The two adversaries differ in the information available to them: the CSP observes protected representations, whereas the curious user additionally holds aligned pairs of original embeddings and protected representations.

**Table 1.** Positioning of this work. For each property, a circle indicates whether it is a stated design objective of the approach (● full, ◑ partial, ○ absent), while ✓ marks the properties evaluated in this work; * indicates an open-source implementation. The circles and checkmarks do not indicate whether a property is actually satisfied; this is determined by our security analysis and experiments. DCPE is included as a search-preserving cryptographic reference, not as an access-control mechanism.

| Approach | Policy isolation | Representation confidentiality | Forgery resistance | Membership hiding |
|---|---|---|---|---|
| Metadata filtering [21, 22] | ● | ○ | ○ | ○ |
| Policy-specific indexes [23]* | ● | ○ | ○ | ○ |
| Cryptographic retrieval [26, 29]* | ● | ● | ○ | ○ |
| DCPE [11, 13]* | ● | ● | ○ | ○ |
| Geometric access control [8, 10]*, [9] | ● | ◑ | ○ | ○ |
| This work (evaluation) | ✓ | ✓ | ✓ | ✓ |

## 3 System and Threat Model

### 3.1 System Model

Figure 1 summarizes the system and threat model. We consider a multi-policy RAG system in which documents governed by different access-control policies are stored in a vector database. A public encoder $\Phi$ maps each document chunk $c_i$ to an embedding $e_i = \Phi(c_i) \in \mathbb{R}^d$. Each policy $p$ is associated with a secret transformation $f_p$. The transformations are applied by a *trusted server* that holds the policy keys, whereas the vector database that stores the results is

untrusted. We assume that embeddings are $L_2$-normalized and that retrieval ranks candidates by cosine similarity, i.e., the inner product of unit-norm vectors. During ingestion, the server applies the corresponding transformation to each embedding:

$$v'_i = f_p(e_i), \tag{1}$$

and stores the resulting protected representation $v'_i$ in the vector database.

At query time, an authenticated user is associated with one or more authorized policies. For a query $q$, the server computes $e_q = \Phi(q)$ and applies the transformation for an authorized policy before submitting the transformed query to the vector database. The retrieved chunks are then returned to a trusted LLM for response generation.

Policy keys never leave the server. The vector database stores only protected representations, and similarity search is performed by a nearest-neighbor engine (e.g., FAISS [7]). The provider can access these representations but never the policy keys.

This work focuses on the security of the protected representations used for retrieval. Document confidentiality at rest and security of the generation component are therefore outside the scope of our analysis.

### 3.2 Adversary Model

We consider honest-but-curious adversaries that follow the retrieval protocol but may analyze protected representations available to them. All adversaries know the public encoder $\Phi$ but do not possess the secret transformation key of the target policy. We distinguish two levels of adversarial knowledge:

- *Curious Cloud Service Provider (CSP).* The provider operating the vector database has access to the protected representations stored in the vector database but no corresponding original embeddings for the target policy and no policy key. This models the information available to an untrusted provider.
- *Curious User.* The adversary is a legitimately enrolled user that is authorized under one policy but not under the target policy. For its own authorized policy, it obtains pairs of original embeddings and protected representations through the normal system interface. For the target policy, we assume that it additionally obtains a bounded set of such pairs through collusion with the CSP; this is the most capable adversary considered in this work. Because the encoder $\Phi$ is public, each such pair, which we call an *aligned pair*, provides an observation:

$$(e_i, v'_i) = (\Phi(c_i), f_p(e_i)). \tag{2}$$

These observations allow the user to estimate the policy-specific transformation without obtaining its secret key. This learnability from aligned pairs is a known weakness of distance-preserving transformations [16, 17]; we examine its implications for both the confidentiality of policy-specific transformations (Section 4.3) and their resistance to query forgery (Section 4.4).

For forgery evaluation, we additionally assume that the curious user can submit a self-constructed protected representation of a query for similarity search under the target policy, without possessing the target policy key or going through the trusted server's normal query transformation. The user therefore attempts to construct a target-policy-compatible query using the transformation estimated from the available aligned observations. This models a compromised or colluding client and is distinct from the passive CSP capability. This capability is not part of normal operation; instead, it characterizes the residual protection of the transformation once informative observations become available.

We do not consider compromise of the trusted server or disclosure of policy keys. Active attacks on the retrieval infrastructure, such as index tampering, denial of service, or malicious document insertion, are outside the scope of this work. Query forgery is evaluated separately as an authorization attack against similarity search.

## 4 Security Analysis

We analyze the security properties of search-preserving geometric access control through a component model of the underlying geometric transformations (Section 4.1) and under the threat model defined in Section 3. We treat *similarity preservation* as a functional requirement: the transformation must preserve the similarity relationships needed for retrieval, which we evaluate through retrieval quality relative to the unprotected embeddings (Section 5). Beyond this requirement, we study four security properties: policy isolation, representation confidentiality, query forgery resistance, and policy-membership hiding. A document containing the complete derivations of this section (`docs/derivations.pdf`) is available in the anonymized artifact.[1]

### 4.1 A Component Model of the Geometric Family

We represent the considered geometric mechanisms through a common component model. The policy-dependent components may include an orthogonal map $R_p$ (a key-dependent coordinate permutation is a particular case, since a permutation matrix is orthogonal), a geometric anchor $a_p \in \mathbb{R}^m$, where $m$ is the anchor-block dimension and $m$ must be at least the number of policies for mutually orthogonal anchors, a bounded perturbation $n$, and a possibly nonlinear map $g$. The resulting protected representation is:

$$v' = N([\, g(R_p e + n)\,;\, M a_p\,])\,, \tag{3}$$

where $M$ scales the anchor and $N(\cdot)$ renormalizes the result to unit length. Here $g$ denotes a possibly nonlinear map that some constructions add to hinder inversion of the protected representation. It is the identity map, denoted $g = \mathrm{id}$, for the linear constructions, whereas Trans-RAG applies a bounded nonlinearity [8].

[1] `https://anonymous.4open.science/r/geoac-rag-repro`

**Table 2.** Mapping of geometric components to the evaluated mechanisms ($R_p$: policy-specific orthogonal map; $Ma_p$: policy anchor; Noise: additive noise or bounded perturbation $n$; Nonlin.: nonlinear map $g$).

| Mechanism | $R_p$ | $Ma_p$ | Noise | Nonlin. |
|---|---|---|---|---|
| Trans-RAG [8] | ✓ | – | ✓ | ✓[†] |
| Subspace rotation [9] | ✓ | – | – | – |
| Key-param. [10] | –[‡] | ✓ | – | – |
| DCPE [11, 12] | ✓ | – | ✓ | – |

[†] Trans-RAG applies $g(x) = \tanh(\beta x)/\beta$ elementwise with $\beta = 0.1$; the complete mechanism is evaluated in Section 5.
[‡] Key-param. [10] applies a rotation shared across all policies (not policy-specific, hence $R_p$ marked absent); only its anchor is policy-dependent.

Only $R_p$ is orthogonal: once $n$, $g$, and the renormalization are applied, the overall transformation $f_p$ is no longer an isometry. The isolation analysis of Section 4.2 therefore takes $g = \mathrm{id}$ and is first stated without $n$; the effect of $g$ and $n$ is evaluated empirically in Section 5.

An authorized query is transformed with the same policy-specific components, without the document-side perturbation:

$$v'_q = N([\, g(R_p e_q)\,;\, Ma_p\,]). \tag{4}$$

We denote the noise-free linear signal-and-anchor component by $u_p(e) = [\, R_p e\,;\, Ma_p\,]$; it is the object of the isolation analysis below.

*Relation to existing approaches.* The component model is not a new access-control mechanism. It is an analytical abstraction of the published constructions (Table 2); all empirical attacks are performed on the published mechanisms and their fusion (Section 5).

### 4.2 Policy Isolation

A query authorized under policy $p$ should retrieve documents governed by $p$ rather than documents belonging to another policy. To quantify residual cross-policy retrieval, we define two complementary rates: the *query violation rate* (QVR), the fraction of queries whose top-$k$ contains at least one unauthorized document (*how many* queries leak), and the *unauthorized chunk rate* (UCR), the fraction of retrieved chunks that are unauthorized (*how much* leaks). Both are zero under perfect isolation.

Under the cosine-similarity retrieval setting of Section 3.1, we derive the conditions for isolation between policies. When a policy anchor is used, as in [10], policy separation follows from its geometric contribution, and we seek the condition on $M$ that ensures every same-policy document ranks above every

cross-policy document, regardless of content. For an authorized query $e_q$ under policy $p$ and a document $e$ governed by policy $p$ or by another policy $p' \neq p$, we assume unit-norm embeddings ($\|e\| = \|e_q\| = 1$), an orthogonal map ($R_p^\top R_p = I$), unit-norm and mutually orthogonal anchors ($\|a_p\| = 1$ and $\langle a_p, a_{p'} \rangle = 0$ for $p \neq p'$), and $g = \mathrm{id}$. The same-policy inner product before normalization is:

$$\begin{aligned} \langle u_p(e_q), u_p(e) \rangle &= \langle R_p e_q, R_p e \rangle + M^2 \langle a_p, a_p \rangle \\ &= \langle e_q, e \rangle + M^2, \end{aligned} \tag{5}$$

where we use $R_p^\top R_p = I$ and $\|a_p\| = 1$, whereas for a cross-policy pair it is:

$$\begin{aligned} \langle u_p(e_q), u_{p'}(e) \rangle &= \langle R_p e_q, R_{p'} e \rangle + M^2 \langle a_p, a_{p'} \rangle \\ &= \langle R_p e_q, R_{p'} e \rangle, \end{aligned} \tag{6}$$

since $\langle a_p, a_{p'} \rangle = 0$ for $p \neq p'$.

For the noise-free idealization, in which each protected representation is $u_p(e)$, the squared norm is:

$$\|u_p(e)\|^2 = \|R_p e\|^2 + M^2 \|a_p\|^2 = 1 + M^2. \tag{7}$$

Similarly, $\|u_p(e_q)\|^2 = 1 + M^2$, so both cosine similarities have the same denominator $1 + M^2$:

$$\mathrm{sim}_{\mathrm{same}} = \frac{\langle e_q, e \rangle + M^2}{1 + M^2}, \tag{8}$$

and,

$$\mathrm{sim}_{\mathrm{cross}} = \frac{\langle R_p e_q, R_{p'} e \rangle}{1 + M^2}. \tag{9}$$

Because the embeddings are normalized, $-1 \leq \langle e_q, e \rangle \leq 1$, while by the Cauchy–Schwarz inequality,

$$|\langle R_p e_q, R_{p'} e \rangle| \leq 1. \tag{10}$$

The common denominator therefore does not affect the relative ranking. The same-policy numerator is at least $M^2 - 1$, and the cross-policy numerator is at most 1 by Eq. (10). Therefore, every same-policy pair ranks above every cross-policy pair if $M^2 - 1 > 1$, which gives:

$$M^2 > 2. \tag{11}$$

The anchor is not strictly necessary for probabilistic separation. With independently keyed, uniformly random orthogonal maps, cross-policy inner products concentrate around zero in high dimension, with fluctuations of order $O(1/\sqrt{d})$, where $d$ is the embedding dimension. The anchor instead provides the deterministic separation guarantee of Eq. (11). The effect of the bounded perturbation is evaluated empirically in Section 5. This establishes the intended retrieval property, but does not imply that the transformation or policy membership is hidden from an observer.

### 4.3 Representation Confidentiality

The original embedding should not be readily recoverable from the protected representation available to an adversary. We measure this property through functional reconstruction resistance: the number of aligned pairs required to reproduce the retrieval behavior of the original embedding (Section 5).

The component model shows that a search-preserving transformation may prevent cross-policy retrieval while remaining learnable from sufficiently informative observations. For the curious user, the aligned observations have the form:

$$(e_i, v_i') = (e_i, f_p(e_i)), \tag{12}$$

where $e_i$ is the original embedding and $v_i'$ is its protected representation. For a linear content-dependent transformation, these pairs constrain the policy-specific mapping. In particular, for an orthogonal transformation $R_p$, sufficiently many linearly independent pairs allow the mapping to be estimated by orthogonal Procrustes alignment [6]. This is analogous to known-plaintext attacks on distance-preserving transformations [16, 17], which we extend here to the RAG access-control setting. Thus, keeping the policy transformation secret does not by itself prevent it from being learned. We therefore evaluate reconstruction cost under different observation budgets in Section 5.

### 4.4 Query Forgery Resistance

Using the policy transformation estimated from aligned pairs (Section 3), the curious user should not be able to construct a query accepted under a policy it is not authorized for. We consider *constructive forgery*: the adversary estimates the target transformation from prior observations and applies it to a fresh query, rather than replaying a captured legitimate query.

Forgery is evaluated separately from reconstruction because it targets a different object: a policy-compatible query rather than an original embedding. The forged query competes against the genuine protected representations of the target policy, so its success depends on the accuracy of the estimated transformation, not on the fidelity of any single reconstructed embedding.

### 4.5 Policy-Membership Hiding

An observer of the vector database should not be able to reliably infer which policy a protected representation belongs to. This property is distinct from policy isolation: a system may prevent unauthorized retrieval while still revealing policy membership through the geometry of its representations. Policy-dependent geometric separation can make membership observable. In particular, policy-specific anchors give representations from the same policy a common geometric component, while policy-specific transformations may create additional patterns that distinguish policies. We therefore evaluate membership inference separately from retrieval isolation, by measuring how reliably an observer can infer the policy of a protected representation.

## 5 Experimental Evaluation

We evaluate the security properties introduced in Section 4 on published search-preserving geometric mechanisms and their fusion, using the authors' own implementations. The evaluation addresses four questions: (i) do the mechanisms preserve retrieval utility while enforcing policy isolation? (ii) how difficult is it to recover original embeddings from protected representations? (iii) how difficult is it to forge a query for a policy the adversary is not authorized for? and (iv) can an observer infer policy membership from the vector database?

### 5.1 Experimental Setup

We evaluate the mechanisms on the datasets and encoders summarized in Table 3. We run three published mechanisms with public implementations: the query-centric transformation of Trans-RAG [8], the key-parametrized dimension augmentation of [10], and DCPE through the IronCore SDK [12]. Using the authors' own implementations ensures that the measurements characterize the published mechanisms rather than a reinterpretation.

We additionally evaluate a *fusion* that applies Trans-RAG's transformation followed by the key-parametrized augmentation of [10], to test whether composing the two search-preserving geometric access-control mechanisms strengthens their security properties. DCPE is evaluated separately as a distance-comparison-preserving encryption scheme and is not part of the fusion.

Policy isolation is evaluated on a balanced MedQuAD sample with attribute-based access-control (ABAC) metadata based on document type, project, and region, defining nine policies. This corpus provides the multi-policy setting needed to measure unauthorized retrieval. Retrieval utility is evaluated separately on MS MARCO [5], a passage-ranking benchmark with real-world search queries, to measure whether the transformations preserve semantic retrieval quality. We additionally evaluate retrieval utility on three BEIR datasets [3] to test whether this behavior extends across different retrieval corpora.

Reconstruction and forgery are evaluated on a designated target-policy subset of MedQuAD containing 3,500 stored embeddings and 300 held-out queries. This controlled single-policy setting isolates the cost of learning the target transformation and using it for reconstruction or query forgery.

Retrieval utility is measured using Recall@10 and nDCG@10, while policy isolation is measured using QVR and UCR (Section 4.2). Attack success is measured functionally, by the overlap between the top-10 results obtained with the estimated and true policy transformations. We denote by $n_\tau^\star$ the smallest number of aligned pairs required to reach an overlap of $\tau$. We use $\tau = 0.8$ as the primary threshold and $\tau = 0.9$ as a stricter secondary threshold. Thus, $n_\tau^\star$ measures the empirical number of observations required to reproduce at least $\tau$ of the reference top-10 results; it is not a cryptographic security guarantee.

Reconstruction and forgery use the same linear least-squares estimator. We fit the policy transformation from the aligned pairs using the Moore–Penrose pseudo-inverse; for the anchor-based mechanism, the constant policy anchor is

**Table 3.** Evaluation corpora, tasks, and sizes. Reported sizes are the indexed corpora and query sets. The MedQuAD ABAC sample is balanced across nine policies (400 documents per policy); the full corpus contains 54,269 documents.

| Corpus | Purpose | Docs | Queries | Policies |
|---|---|---|---|---|
| MS MARCO [5] | Retrieval utility | 3,047 | 306 | none |
| BEIR NFCorpus [3] | Retrieval utility | 3,633 | 323 | none |
| BEIR SciFact [3] | Retrieval utility | 5,183 | 300 | none |
| BEIR FiQA [3] | Retrieval utility | 57,638 | 648 | none |
| MedQuAD ABAC sample [4] | Policy isolation | 3,600 | 360 | 9 |
| MedQuAD target-policy subset | Reconstruction, forgery | 3,500 | 300 | 1 |

Encoders: MiniLM ($d$=384, primary), MPNet ($d$=768), GTE-large ($d$=1024).

recovered from a single pair. We use the unconstrained least-squares solution rather than an orthogonality-constrained one, so that it also accommodates DCPE's global scaling while remaining robust to its additive perturbation. The attack cost $n_\tau^\star$ is measured on a single witness sample using a coarse pair grid, with 100-pair increments from 100 to 600 and larger steps up to 3,500. Consequently, differences such as 200 versus 300 pairs should not be interpreted as precise one-step estimates. Isolation experiments are repeated with three random seeds. All experiments run on a Linux server (Ubuntu 24.04) with an Intel Xeon Gold 5416S CPU and an NVIDIA L40 GPU. Embeddings are encoded on the GPU, while attack-cost measurements use a single CPU thread.

### 5.2 Does Geometric Transformation Preserve Retrieval and Isolation?

Without policy-specific geometry, the baseline provides no policy isolation. Every query returns at least one unauthorized document (QVR = 1.00), while about 8–9 of the 10 returned chunks are unauthorized (UCR $\approx$ 0.86). Across the evaluated geometric configurations, both QVR and UCR are zero, with zero variance across seeds.

On the policy-isolation corpus, geometric isolation raises authorized within-document Recall@10 from 0.28 for the unfiltered baseline to 0.44, matching an oracle pre-filter that removes unauthorized documents before ranking (0.44). Policy isolation is evaluated only on the policy-labeled MedQuAD corpus. Retrieval quality is likewise preserved across the three BEIR collections and the three encoders: Recall@10 stays within 0.01 of the unprotected reference in all nine dataset–encoder combinations.

Consistent with the analysis of Section 4.2, independently keyed per-policy orthogonal maps (the rotation of Trans-RAG [8] and the key-dependent coordinate permutation of DCPE [11]) already produce zero measured cross-policy retrieval, while the anchor of the key-parametrized augmentation [10] yields the same isolation (Table 4).

For retrieval utility on MS MARCO, the unprotected reference system achieves Recall@10 = 0.986 and nDCG@10 = 0.699. The geometric transforms preserve retrieval: a noise-free rotation or anchor is order-preserving, while mechanisms that add a per-vector perturbation (Trans-RAG's blinding noise and DCPE's perturbation) affect the ranking only marginally, keeping the top-10 agreement with the unprotected ranking at R@10 $\geq$ 0.91 across the three encoders (Table 4).

*Metric sensitivity of dimension augmentation.* The key-parametrized augmentation of [10] adds a large-valued coordinate per policy, whose magnitude is calibrated for a specific similarity metric. With the article's $\alpha = 10^3$ under cosine similarity, or the code's $\alpha = 10^9$ under the Euclidean metric it targets, retrieval is preserved (Recall@10 $\geq$ 0.98 across the three encoders). Porting the code's $\alpha = 10^9$ to a cosine-based store, the common RAG configuration, collapses retrieval (Recall@10 = 0.02) while isolation still holds, because, with $M = \alpha$ in Eq. (8), a difference $\Delta\cos$ between two underlying cosine scores becomes $\Delta\cos/(1 + M^2)$ and can fall below floating-point resolution, making the resulting similarities numerically indistinguishable. The mechanism is therefore metric-dependent, which is a portability caveat rather than a flaw.

### 5.3 Can Protected Representations Be Reconstructed?

Table 4 reports the reconstruction cost for the curious user on the three published mechanisms and their fusion, run on the authors' own code across three encoders of increasing dimension. The same table also reports forgery and membership results, discussed in Sections 5.4 and 5.5.

The reconstruction results show two main observations. First, across the three encoders, increasing the encoder dimension from $d = 384$ to $d = 1024$ does not increase the observed reconstruction cost: all evaluated transformations are reconstructible within 200–400 aligned pairs (Figure 2). This is consistent with the relatively low effective dimensionality of the embedding spaces: the participation ratio, which summarizes how many directions carry most of the variance, is 54, 53, and 64 for the three encoders, while 90% of the variance is captured by 159, 182, and 291 directions, respectively. The larger encoder therefore does not raise the observed reconstruction budget; the unprotected embeddings are likewise recovered within 300 pairs.

Second, added noise or nonlinearity limits the *fidelity* of reconstruction but not its *cost*. Trans-RAG's blinding noise and nonlinearity [8], and the bounded perturbation of DCPE [11], cap the top-10 overlap below one (0.86–0.94), whereas the key-parametrized augmentation [10], being a constant per-policy coordinate, leaves reconstruction of the rotated content unimpeded (ceiling 1.00). Yet all reach the operational success threshold within a comparable pair budget. Policy-specific rotations and anchors therefore do not increase the observed reconstruction cost.

### 5.4 Can the Same Observations Forge Policy-Compatible Queries?

The aligned pairs used for reconstruction also let the curious user estimate the policy transformation and forge a query under a policy it is not authorized for.

**Table 4.** Published mechanisms and fusion across encoder dimensions. Reconstruction and forgery report $n^\star_{0.8}$ and the overlap ceiling at the largest budget. R@10 is the same-policy retrieval overlap between protected and unprotected top-10 results; ARI measures policy-membership clustering by the curious CSP. Isolation is ✓ when QVR=UCR≈0. Shaded cells mark a failed property: isolation if QVR $> 0$; confidentiality and forgery resistance if the attack ceiling reaches $\tau = 0.8$; membership hiding if ARI $\geq 0.5$. Pair counts $n^\star_{0.8}$ are costs, not verdicts, and are left unshaded.

| | | | Reconstruction | | Forgery | | | |
|---|---|---|---|---|---|---|---|---|
| Mechanism | $d$ | Iso. | $n^\star_{0.8}$ | ceil. | $n^\star_{0.8}$ | ceil. | R@10 | ARI |
| No access control | 384 | ✗ | 300 | 1.00 | n/a | n/a | 1.00 | 0.00 |
| Trans-RAG [8] | 384 | ✓ | 300 | 0.86 | 500 | 0.86 | 0.93 | 1.00 |
| | 768 | ✓ | 300 | 0.93 | 1,000 | 0.90 | 0.95 | 1.00 |
| | 1,024 | ✓ | 400 | 0.87 | 1,800 | 0.84 | 0.91 | 1.00 |
| Key-param. aug. [10] | 384 | ✓ | 300 | 1.00 | 300 | 0.98 | 1.00 | 1.00 |
| | 768 | ✓ | 200 | 1.00 | 200 | 1.00 | 1.00 | 1.00 |
| | 1,024 | ✓ | 300 | 1.00 | 300 | 1.00 | 1.00 | 1.00 |
| DCPE [11, 12] | 384 | ✓ | 300 | 0.90 | 500 | 0.90 | 0.97 | 0.97 |
| | 768 | ✓ | 300 | 0.92 | 1,000 | 0.91 | 0.97 | 1.00 |
| | 1,024 | ✓ | 400 | 0.91 | 1,200 | 0.90 | 0.96 | 1.00 |
| Fusion (TR⊕aug.) | 384 | ✓ | 200 | 0.86 | 500 | 0.86 | 0.93 | 1.00 |
| | 768 | ✓ | 200 | 0.94 | 1,000 | 0.90 | 0.95 | 1.00 |
| | 1,024 | ✓ | 400 | 0.88 | 1,800 | 0.84 | 0.91 | 1.00 |

Table 4 reports forgery cost alongside reconstruction. Forgery reaches 0.84–1.00 of the legitimate top-10 overlap across the three published mechanisms and their fusion, so none resists the attack. Its cost, however, differs from reconstruction in how it scales with encoder dimension. For mechanisms whose target-policy transformation is secret (Trans-RAG, DCPE, and the Trans-RAG⊕augmentation fusion), the attacker must estimate the full target transformation from the aligned pairs. Forgery cost therefore grows with encoder dimension, from 500 to 1,800 pairs as $d$ increases from 384 to 1024 (Figure 2).

The key-parametrized augmentation [10] behaves differently: it applies a fixed rotation shared across all policies and separates policies only by a constant per-policy anchor. Because the same rotation appears in every policy's pairs, an adversary enrolled in any policy can estimate it from its own authorized pairs, while only the low-dimensional anchor remains target-specific. The reported $n^\star$ therefore counts the pairs needed from the adversary's own policy; for the target policy, a single pair suffices to recover the target anchor (Section 5.1). Forgery thus does not scale with $d$ and stays at 200–300 pairs, the same cost as reconstruction. For the mechanisms with a secret target-policy transformation, forgery costs more than reconstruction (500 versus 300 pairs at $d = 384$). This higher cost is consistent with the stricter requirement imposed by forgery: the

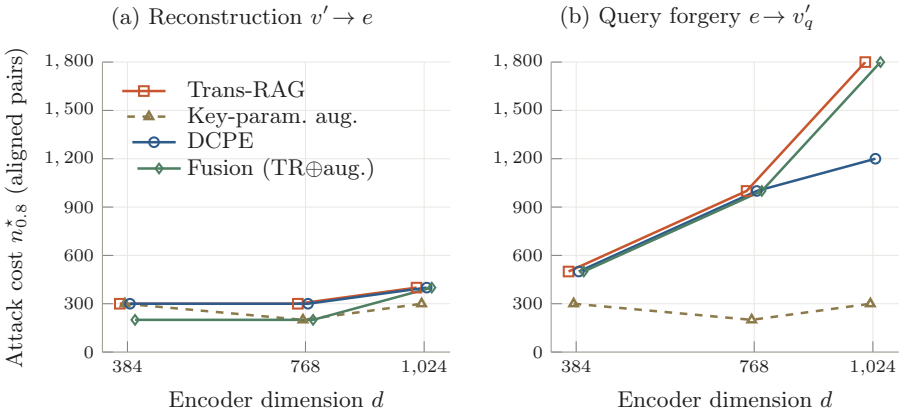


**Fig. 2.** Attack cost $n^\star_{0.8}$ versus encoder dimension, with one line per mechanism (values from Table 4; the same symbol denotes a mechanism in both panels). (a) Reconstruction stays within 200–400 aligned pairs as $d$ grows from 384 to 1024. (b) Query forgery grows with $d$ for mechanisms whose per-policy transformation is secret (Trans-RAG, DCPE, and their fusion), reaching 1,200–1,800 pairs, while the key-parametrized augmentation, whose rotation is shared across policies, stays at 200–300. Markers are slightly horizontally offset where values coincide.

estimated transformation must produce a query that retrieves the target policy's genuine protected representations.

### 5.5 Does Geometric Isolation Leak Policy Membership?

The curious CSP clusters the stored protected representations using $k$-means, without any aligned pair or policy key. The Adjusted Rand Index (ARI) measures agreement between the inferred clusters and the true policy partition. Across the three encoders, Trans-RAG, the key-parametrized augmentation [10], and their fusion reach ARI = 1.00, while DCPE reaches ARI = 0.97–1.00, compared with ARI $\approx$ 0.001 for the unprotected baseline (Table 4). Thus, policy membership is highly recoverable even without aligned pairs or policy keys, including for DCPE, a cryptographic distance-comparison-preserving scheme.

This exposure is consistent with embedding anisotropy. Learned embeddings concentrate around a common mean, so applying a distinct per-policy key produces distinct per-policy geometry, whether through a rotated or permuted mean for the orthogonal-map mechanisms or through an added coordinate for the anchor. These differences remain separable by unsupervised clustering. The same geometric separation that supports policy isolation can therefore expose policy membership even when unauthorized retrieval is prevented.

### 5.6 Summary of Findings

Across the evaluated mechanisms, policy isolation does not imply reconstruction resistance, forgery resistance, or policy-membership hiding. All geometric mechanisms achieve the measured isolation, yet policy membership remains recoverable from the protected representations. Increasing encoder dimension does not increase the observed reconstruction cost, while forgery cost increases with dimension for mechanisms with a target-policy-specific transformation. Adding noise or nonlinearity limits reconstruction fidelity but not the observed pair budget, and the evaluated fusion does not increase the attack costs.

The unprotected baseline provides a complementary reference: it fails isolation (QVR=1.00/UCR=0.86) but does not expose policy membership (ARI=0.00). The geometric mechanisms reverse these two properties: they provide measured isolation while exposing policy membership.

## 6 Discussion

The results clarify the security boundary of search-preserving geometric access control. Geometric transformations can enforce policy separation directly in the embedding space, but policy isolation does not imply confidentiality, forgery resistance, or policy-membership hiding. Once the curious user holds aligned pairs, the evaluated mechanisms remain learnable for both reconstruction and query forgery, while their policy-dependent geometry can expose membership information.

### 6.1 Limitations and Security Boundary

The reconstruction and forgery results assume that the curious user can obtain aligned pairs of original embeddings and protected representations for the target policy. This provides more information than is normally available to the CSP and is therefore best understood as a test of the residual protection once informative observations are available, rather than as an assumption that such observations are always available in practice. The evaluation covers the published mechanisms considered in this work and their fusion, not the full space of search-preserving constructions. The measured attack costs therefore characterize these mechanisms under the evaluated threat model and do not constitute cryptographic security guarantees for the broader paradigm. Our formal analysis of isolation takes $g = \mathrm{id}$; the nonlinearity of Trans-RAG is retained in the empirical evaluation but is not modeled analytically. Overall, the results support treating policy isolation as a distinct security property rather than as a proxy for broader confidentiality or anti-forgery guarantees. The required protection therefore depends on the security objective and the adversarial capabilities being considered.

### 6.2 Towards Mitigations

The results also suggest several directions for mitigation. First, query authorization should not rely solely on the geometric transformation: a trusted server

could authenticate transformed queries before they reach the vector database, preventing a user who has learned the transformation from directly submitting forged queries. Second, limiting the number of observations available under a single policy transformation, for example through periodic re-keying or stricter separation of protected representations, may reduce the effectiveness of attacks based on aligned pairs. These measures address different parts of the problem and should be evaluated explicitly in future work. When representation confidentiality is a primary requirement, cryptographic retrieval mechanisms may also be preferable to relying on geometric separation alone.

## 7 Conclusion

Search-preserving geometric access control can provide effective policy isolation by embedding policy separation into the embedding space. Our analysis and experiments show that this isolation does not imply representation confidentiality, query forgery resistance, or policy-membership hiding. With aligned pairs of original embeddings and protected representations, a curious user can recover those embeddings from a few hundred observations and forge policy-compatible queries at a higher, dimension-dependent cost for secret per-policy transformations. Composing published mechanisms does not substantially increase these costs. These results show that search-preserving geometric access control should be evaluated beyond retrieval effectiveness and policy isolation. Future work will evaluate mitigation strategies and extend the analysis to broader classes of search-preserving constructions.

*Ethical statement.* This work evaluates search-preserving geometric access-control mechanisms using controlled experiments, publicly available data, and implementations released as part of scientific publications. Our evaluation is limited to research implementations intended as proofs of concept; to our knowledge, none of the implementations evaluated in this work is known to be deployed as part of a commercial product. The attacks target experimental representations and retrieval interfaces in these research implementations. No real-world system was compromised and no sensitive user data were accessed. In the event of acceptance, we plan to contact the authors of the evaluated implementations following a responsible-disclosure process before the final publication of the work. The reproducibility artifact is described below.

*Open Science.* All code, dataset-preparation scripts, and configurations required to reproduce the reported results, including the corpus statistics of Table 3 and the reconstruction, forgery, and membership costs of Table 4, are available as an anonymized artifact: `https://anonymous.4open.science/r/geoac-rag-repro`. The repository provides scripts for reproducing the main experiments, generating the reported tables and figures, and evaluating the considered geometric mechanisms and adversarial procedures. The artifact also contains a document with the complete derivations of Section 4. Long-term archival storage is provided through Zenodo.

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